

Ants and their larvae shelter inside the swollen thorn of an East African whistling thorn acacia tree. In return the ants swarm from their nests to protect the tree from herbivores.

Benefits for all

There are two fundamental types of mutualism—service-resource and service-service relationships. They are defined by the nature of the relationship between the partner organisms, whether it is the provision of a service or the supply of a resource—both are usually key to survival. Service-resource relationships are common in nature, with the fertilization, or pollination, of flowers by butterflies, moths, bees, flies, wasps, beetles, bats, or birds the most widespread example. The resource (pollen) is provided by the flower, and the service (pollination) is provided by the animal. It is estimated that nearly three-quarters of flowering plants (some